

Q. What is Linear Combination?

$\Rightarrow$ Let $v_1, v_2, v_3, \dots, v_p \in \mathbb{R}^n$ and $a_1, a_2, a_3, \dots, a_p$ are scalars then a vector 'b' is called a linear combination of $v_1, v_2, \dots, v_p$ if b can be written as

$$b = a_1 v_1 + a_2 v_2 + \dots + a_p v_p$$

$$AX = b$$

$$[v_1, v_2, \dots, v_p] \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = b$$

$$a_1 v_1 + a_2 v_2 + \dots + a_p v_p = b$$

$a_1, a_2, \dots, a_p$ normally called weights

Example:

Show that $v = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ is a

Linear Combination of $\begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Sol:

Let a, b, c are scalar

$$\vec{V} = a \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} + b \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix} + c \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 3a \\ 0 \\ -a \end{bmatrix} + \begin{bmatrix} b \\ 4b \\ 3b \end{bmatrix} + \begin{bmatrix} -c \\ c \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 3a + b - c \\ 4b + c \\ -a + 3b \end{bmatrix}$$

$$3a + b - c = 4 \quad \text{--- (i)}$$

$$4b + c = 3 \quad \text{--- (ii)}$$

$$-a + 3b = 7 \quad \text{--- (iii)}$$

multiply eq (iii) by 3

$$-3a + 9b = 21 \quad \text{--- (iv)}$$

Add (i) & (iv)

$$-3a + 9b = 21$$

$$3a + b - c = 4$$

$$10b - c = 25 \quad \text{--- (v)}$$

Add (ii) & (v)

$$10b - c = 25$$

$$4b + c = 3$$

$$14b = 28$$

$$\Rightarrow \boxed{b = 2}$$

Put $b=2$ in (ii)

$$4(2) + c = 3$$

$$8 + c = 3 \Rightarrow \boxed{c = -5}$$

$$3a + b - c = 4$$

$$3a + 2 - (-5) = 4$$

$$3a + 2 + 5 = 4$$

$$3a = -3$$

$$\boxed{a = -1}$$

$$\begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = -1 \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix} - 5 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Example #2:

if possible, write $\vec{v}$:

$\vec{v} = (-1, 7, 2)$ as a linear

combination of $\vec{u} = (1, 3, 5)$,

$\vec{v}_2 = (2, -1, 3)$ & $\vec{v}_3 = (-3, 2, -4)$

Let a, b, c are scalar

$$\vec{v} = a\vec{u} + b\vec{v}_2 + c\vec{v}_3$$

$$(-1, 7, 2) = a(1, 3, 5) + b(2, -1, 3) + c(-3, 2, -4)$$

$$(-1, 7, 2) = (a, 3a, 5a) + (2b, -b, 3b) + (-3c, 2c, -4c)$$

$$(-1, 7, 2) = (a + 2b - 3c, 3a - b + 2c, 5a + 3b - 4c)$$

$$a + 2b - 3c = -1$$

$$3a + b + 2c = 7$$

$$5a + 3b - 4c = 2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 3 & -1 & 2 & 7 \\ 5 & 3 & -4 & 2 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1/7 & 0 \\ 0 & 1 & 11/7 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

$\Rightarrow$ it is impossible, v cannot be written as linear combination of v_1, v_2, v_3 .

D.

Q# Express the vector $6+11x+6x^2$
as a linear combination
of (PP-2020)

$$P_1 = 2+x+4x^2$$

$$P_2 = 1-x+3x^2$$

$$P_3 = 3+2x+5x^2$$

$$\vec{V} = \vec{V_1}a + \vec{V_2}b + \vec{V_3}c$$

$$\Rightarrow 6+11x+6x^2 = a(2+x+4x^2) + b(1-x+3x^2) + c(3+2x+5x^2)$$

$$\Rightarrow 6+11x+6x^2 = (2a+b+3c) + (a-b+2c)x + (4a+3b+5c)x^2$$

$$2a+b+3c=6$$

$$a-b+2c=11$$

$$4a+3b+5c=6$$

$$\left[\begin{array}{ccc|c} 2 & 1 & 3 & 6 \\ 1 & -1 & 2 & 11 \\ 4 & 3 & 5 & 6 \end{array} \right]$$

$$R_2 \leftrightarrow R_1 \left[\begin{array}{ccc|c} 1 & -1 & 2 & 11 \\ 2 & 1 & 3 & 6 \\ 4 & 3 & 5 & 6 \end{array} \right]$$

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$$\begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \end{array} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 11 \\ 0 & 3 & -1 & -16 \\ 0 & 7 & -3 & -32 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2 \left[\begin{array}{ccc|c} 1 & -1 & 2 & 11 \\ 0 & 3 & -1 & -16 \\ 0 & 4 & -2 & -22 \end{array} \right]$$

$$\frac{-1}{2} R_3 \rightarrow R_3 \left[\begin{array}{ccc|c} 1 & -1 & 2 & 11 \\ 0 & 3 & -1 & -16 \\ 0 & -2 & 1 & 11 \end{array} \right]$$

$$\begin{array}{l} R_2 \rightarrow R_2 + R_3 \\ R_3 \rightarrow R_3 + 2R_2 \end{array} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 11 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

$$R_1 \rightarrow R_1 + R_2 \left[\begin{array}{ccc|c} 1 & 0 & 2 & 6 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

$$R_1 \rightarrow R_1 - 2R_3 \left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

$$a = 4$$

$$b = -5$$

$$c = 1$$

$$6 + 11x + 6x^2 = 4(2 + x + 4x^2) - 5(1 - x + 3x^2) + (3 + 2x + 5x^2)$$

Verification

$$6 + 11x + 6x^2 = (2(4) - 5 + 3) + (4 + 5 + 2)x + (16 - 15 + 5)x^2$$

$$6 + 11x + 6x^2 = 6 + 11x + 6x^2$$

Q# For what values of a is the vector $(a^2, a, 1)$ in $\text{Span}\{(1, 2, 3), (1, 1, 1), (0, 1, 2)\}$ (PP-20/63 2014)

$$\begin{bmatrix} a^2 \\ a \\ 1 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 & : & a^2 \\ 2 & 1 & 1 & : & a \\ 3 & 1 & 2 & : & 1 \end{bmatrix}$$

Original & Successive

$$R_2 \rightarrow R_2 - R_1 \quad \left[\begin{array}{ccc|c} 1 & 1 & 0 & a \\ 2 & 1 & 1 & a \\ 1 & 0 & 1 & 1-a \end{array} \right]$$

DIFFERENTIAL
SPERMAT
CALC

$$R_2 \rightarrow R_2 - 2R_1 \quad \left[\begin{array}{ccc|c} 1 & 1 & 0 & a \\ 0 & -1 & 1 & a-2a \\ 1 & 0 & 1 & 1-a \end{array} \right]$$

$$R_1 \rightarrow R_1 + R_2 \quad \left[\begin{array}{ccc|c} 1 & 0 & 1 & -a+a \\ 0 & -1 & 1 & a-2a \\ 1 & 0 & 1 & 1-a \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_1 \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & -a^2+a-1 \\ 0 & -1 & 1 & a-2a \\ 1 & 0 & 1 & 1-a \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_1 \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & -a^2+a-1 \\ 0 & -1 & 1 & a-2a \\ 0 & 0 & 0 & a^2-2a+1 \end{array} \right]$$

$$a^2 - 2a + 1 = 0$$

$$a^2 - a - a + 1 = 0$$

$$a(a-1) - 1(a-1) = 0$$

$$(a-1)(a-1) = 0$$

$$a-1 = 0$$

$$\begin{cases} x + z = a^2 + a \\ -y + z = a - 2a^2 \end{cases}$$

$$x + z = 1/2$$

$$y + z = 2$$

$$-y + z = 1 - 2(1)$$

$$-y + z = -1$$

for $a = 2$ the vector (a^2, a)

$(1, 1, 1)$ is span $\{(1, 2), (1, 1), (2, 1)\}$

Q# Express $(6, 11, 6)$ as
linear combination of
 $U = (2, 6, 4)$, $V = (1, -1, 3)$ and
 $W = (3, 2, 5)$. (2016-PP)

$$V = u_a + v_b + w_c$$

$$(6, 11, 6) = a(2, 6, 4) + b(1, -1, 3) + c(3, 2, 5)$$

$$\begin{bmatrix} 6 \\ 11 \\ 6 \end{bmatrix} = a \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} + c \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 3 & : & 6 \\ 1 & -1 & 2 & : & 11 \\ 4 & 3 & 5 & : & 6 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2 \begin{bmatrix} 1 & -1 & 2 & : & 11 \\ 2 & 1 & 3 & : & 6 \\ 4 & 3 & 5 & : & 6 \end{bmatrix}$$

$$\begin{aligned} R_2 &\rightarrow R_2 - 2R_1 \\ R_3 &\rightarrow R_3 - 4R_1 \end{aligned} \begin{bmatrix} 1 & -1 & 2 & : & 11 \\ 0 & 3 & -1 & : & -16 \\ 0 & 7 & -3 & : & -38 \end{bmatrix}$$

$$R_1 \rightarrow R_1 + R_2 \begin{bmatrix} 1 & 2 & 1 & : & -5 \\ 0 & 3 & -1 & : & -16 \\ 0 & 7 & -3 & : & -38 \end{bmatrix}$$

$$R_1 \rightarrow R_1 + R_2 \begin{bmatrix} 1 & 5 & 0 & : & -21 \\ 0 & 3 & -1 & : & -16 \\ 0 & 7 & -3 & : & -38 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \begin{bmatrix} 1 & 5 & 0 & : & -21 \\ 0 & 3 & -1 & : & -16 \\ 0 & 4 & -2 & : & -22 \end{bmatrix}$$

$$\frac{1}{2}R_3 \rightarrow R_3 \begin{bmatrix} 1 & 5 & 0 & : & -21 \\ 0 & 3 & -1 & : & -16 \\ 0 & 2 & -1 & : & -11 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_3 \begin{bmatrix} 1 & 5 & 0 & : & -21 \\ 0 & 1 & 0 & : & -5 \\ 0 & 2 & -1 & : & -11 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2 \begin{bmatrix} 1 & 5 & 0 & : & -21 \\ 0 & 1 & 0 & : & -5 \\ 0 & 0 & -1 & : & -1 \end{bmatrix}$$

$$(-1)R_3 \rightarrow R_3 \begin{bmatrix} 1 & 5 & 0 & : & -21 \\ 0 & 1 & 0 & : & -5 \\ 0 & 0 & 1 & : & 1 \end{bmatrix}$$

$$a + 5b = -21$$

$$\boxed{b = -5}$$

$$\boxed{c = 1}$$

$$a + 5(-5) = -21$$

$$\boxed{a = 4}$$

verification

$$\begin{bmatrix} 6 \\ 11 \\ 6 \end{bmatrix} = 4 \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} - 5 \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} + 1 \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 6 \\ 11 \\ 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \\ 16 \end{bmatrix} - \begin{bmatrix} 5 \\ -5 \\ 15 \end{bmatrix} + \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 9 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 6 \\ 11 \\ 6 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \\ 6 \end{bmatrix}$$

Q# Determine whether the vector $v = (3, 3, -4)$ is a Linear Combination of $x = (1, 2, 3)$, $y = (2, 3, 7)$, $z = (3, 5, 6)$ (PP-2021)

$$\underline{v} = x \underline{v}_1 + y \underline{v}_2 + z \underline{v}_3$$

$$\begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} x + \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix} y + \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix} z$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 2 & 3 & 5 & 3 \\ 3 & 7 & 6 & -4 \end{array} \right]$$

$$\begin{array}{l} R_3 \rightarrow R_3 - 3R_1 \\ R_2 \rightarrow R_2 - 2R_1 \end{array} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & -1 & -1 & -3 \\ 0 & 1 & -3 & -13 \end{array} \right]$$

$$R_3 \rightarrow R_3 + R_2 \left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & -4 & -16 \end{array} \right]$$

$$\frac{-1}{4} R_3 \rightarrow R_3 \left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & 1 & 4 \end{array} \right]$$

$$R \quad \boxed{x_3 = -4}$$

$$y = 7 + 4 + 3$$

$$y = 14$$

$$\boxed{y = 14}$$

$$x_1 + 2x_2 + 3x_3 = 3$$

$$x_1 + 2(-1) + 3(1) = 3$$

$$x_1 - 2 + 3 = 3$$

$$x_1 + 10 = 3$$

$$\boxed{x_1 = -7}$$

Verification

$$\begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix} = -7 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + 4 \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} -7 \\ -14 \\ -21 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 12 \\ 20 \\ 24 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$$

Write the vector $v = (1, -2, 5)$ as a linear combination of $v_1 = (1, 1, 1)$, $v_2 = (1, 2, 3)$ and $v_3 = (2, -1, 1)$. (PP-2017)

$$\underline{v} = \underline{v_1}x + \underline{v_2}y + \underline{v_3}z$$

$$\begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}x + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}y + \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}z$$

$$\begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}x + y \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + z \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 1 & 2 & -1 & | & -2 \\ 1 & 3 & 1 & | & 5 \end{bmatrix}$$

$$\begin{array}{l} \sim R_3 \rightarrow R_3 - R_1 \\ \sim R_2 \rightarrow R_2 - R_1 \end{array} \begin{bmatrix} 1 & 1 & 2 & : & 1 \\ 0 & 1 & -3 & : & -3 \\ 0 & 2 & -1 & : & 4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \quad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 1 & 2 & 7 \end{array} \right]$$

$$R_3 \rightarrow R_3 + R_2 \quad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 2 & -1 & 4 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2 \quad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 1 & 2 & 7 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2 \quad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & 5 & 10 \end{array} \right]$$

$$\frac{1}{5} R_3 \quad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & 1 & 2 \end{array} \right] \checkmark$$

$$R_2 \rightarrow R_2 + 3R_3 \quad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & 0 & 3 \frac{1}{5} \\ 0 & 0 & 1 & 2 \end{array} \right] \times$$

$$x = 2$$

$$y - 3z = -3$$

$$y - 6 = -3$$

$$y = -3 + 6$$

$$y = 3$$

$$x + y + 2z = 1$$

$$x + 3 + 2(2) = 1$$

$$x = 1 - 7$$

$$x = -6$$

Verification:

$$\begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} = (-6) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -6 + 3 + 4 \\ -6 + 6 - 2 \\ -6 + 9 + 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix}$$

Given the matrix
 $A = \begin{bmatrix} 5 & 1 \\ -1 & 9 \end{bmatrix}$ as a

Linear Combination of
 $A_1 = \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix}$, $A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$
 and $A_3 = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$.

Let

$$A = k_1 A_1 + k_2 A_2 + k_3 A_3$$

$$\begin{bmatrix} 5 & 1 \\ -1 & 9 \end{bmatrix} = k_1 \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix} + k_2 \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} + k_3 \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 1 \\ -1 & 9 \end{bmatrix} = \begin{bmatrix} k_1 + k_2 + 2k_3 & -k_1 + k_2 + 2k_3 \\ -k_3 & 3k_1 + 2k_2 + k_3 \end{bmatrix}$$

$$\therefore k_1 + k_2 + 2k_3 = 5$$

$$\therefore -k_1 + k_2 + 2k_3 = 1$$

$$\therefore -k_3 = 1$$

$$\therefore 3k_1 + 2k_2 + k_3 = 9$$

The augmented matrix will be

$$\begin{bmatrix} 1 & 1 & 2 & : & 5 \\ 0 & 0 & -1 & : & -1 \\ 3 & 2 & 1 & : & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & : & 5 \\ 0 & 2 & 4 & : & 6 \\ 0 & 0 & -1 & : & -1 \\ 0 & -1 & 5 & : & -6 \end{bmatrix} \begin{array}{l} R_2 + R_1 \\ R_4 - 3R_1 \end{array}$$

$$\frac{R_2}{2} \begin{bmatrix} 1 & 1 & 2 & : & 5 \\ 0 & 1 & 2 & : & 3 \\ 0 & 0 & -1 & : & -1 \\ 0 & -1 & 5 & : & -6 \end{bmatrix}$$

$$R_1 + R_4 \begin{bmatrix} 1 & 1 & 2 & : & 5 \\ 0 & 1 & 2 & : & 3 \\ 0 & 0 & -1 & : & -1 \\ 0 & 0 & -3 & : & -3 \end{bmatrix}$$

The system is inconsistent. Therefore, for a given matrix A is not the linear combination of all three matrices A_1, A_2, A_3 .

Examples:

Show that $\mathbb{R}^n$ is a vector space.

$\Rightarrow$ Let $U = (u_1, u_2, \dots, u_n)$ and $V = (v_1, v_2, \dots, v_n)$ are two vectors in $\mathbb{R}^n$

$\Rightarrow$ The sum $U+V$ is defined by

$$U+V = (u_1+v_1, u_2+v_2, \dots, u_n+v_n)$$

and if k is any scalar, the scalar multiple kU is defined by

$$kU = (ku_1, ku_2, \dots, ku_n)$$

$\Rightarrow$ The operations of addition and scalar multiplication are called the standard operations on $\mathbb{R}^n$.

Show that the set V of all 2×2 matrices with real entries is a vector space.

Let

$$U = \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix}$$

$$\text{and } V = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix}$$

i) $U+V$

$$\begin{aligned} U+V &= \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix} + \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix} \\ &= \begin{bmatrix} U_{11}+V_{11} & U_{12}+V_{12} \\ U_{21}+V_{21} & U_{22}+V_{22} \end{bmatrix} \end{aligned}$$

ii) KU

$$\begin{aligned} KU &= K \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix} \\ &= \begin{bmatrix} KU_{11} & KU_{12} \\ KU_{21} & KU_{22} \end{bmatrix} \end{aligned}$$

iii) $0+U=U+0=U$

$$\begin{aligned} 0+U &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix} \\ &= \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix} \\ &= U \end{aligned}$$

iv) $U+(-U)$

$$= \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix} + \begin{bmatrix} -U_{11} & -U_{12} \\ -U_{21} & -U_{22} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

v) $1U$

$$\begin{aligned} 1U &= 1 \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix} \\ &= \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix} \end{aligned}$$

Vector Space of real-valued functions.

$\Rightarrow$ Let V be the set of real-valued functions defined on the entire real line $(-\infty, +\infty)$

$\Rightarrow$ if f and g are two such functions and k is any real number then

$$(f+g)(x) = f(x) + g(x)$$

and

$$(kf)(x) = k f(x)$$

This vector space is denoted by $F(-\infty, +\infty)$

Vector Space of P_n .

$\Rightarrow$ Let P_n denote the set of all polynomials of degree less than or equal to n .

$\Rightarrow$ if p and q are two polynomials in P_n and k is any real number then

$$(p+q)(x) = p(x) + q(x)$$

$$(kp)(x) = k(p(x)) \quad \text{for all real numbers } x.$$

A Set that is not a vector space.

$\Rightarrow$ Let $V = \mathbb{R}^2$ and define addition and scalar multiplication operations as follows:

if $u = (u_1, u_2)$ and $v = (v_1, v_2)$, then define

$$u+v = (u_1+v_1, u_2+v_2)$$

and if k is any real number, then define

$$ku = (ku_1, 0)$$

for example

$$u = (2, 4), v = (-3, 5) \quad \text{and } k = 7$$

then

$$u+v = (2+(-3), 4+5) = (-1, 9)$$

$$ku = 7u = (7 \times 2, 0) = (14, 0)$$

if $u = (u_1, u_2)$ is such that $u_2 \neq 0$, then

$$1u = 1(u_1, u_2) = (1 \cdot u_1, 0) = (u_1, 0) \neq u$$

$\Rightarrow$ Thus V is not a vector space with the stated operations.

Q# What is Solution Space?

Solution Space:-

Let $Ax=0$ be a homogeneous linear system of m equations in unknowns. Define the set of solution vectors

$$W = \{x \in \mathbb{R}^n : Ax=0\}$$

Then W is called solution space. W is a subspace of $\mathbb{R}^n$.

Q#: What is Space Spanned?

Space Spanned:-

Let $S = \{v_1, v_2, \dots, v_k\}$ be a set of vectors in a vector space V . Then the subspace W of V containing of all linear combination of the vectors in S is called Space spanned by $v_1, v_2, \dots, v_k$ and we say that the vectors $v_1, v_2, \dots, v_k$ span W . we write

$$W = \text{Span}(S)$$

$$\text{or } W = \text{Span}\{v_1, v_2, \dots, v_k\}$$

$$L(S) = \left\{ \sum_{i=1}^n a_i v_i : a_i \in \mathbb{R}, v_i \in S \right\}$$

example:

$$\mathbb{R}^2 = \text{Span}(S) \quad \text{and} \quad \mathbb{R}^3 = \text{Span}\{i, j, k\}$$

e.g

Determine whether $v_1 = (1, 1)$ and $v_2 = (-1, 2)$ span the vectors in space $\mathbb{R}^2$.

Let an arbitrary vector $(a, b) \in \mathbb{R}^2$

$$(a, b) = x(-1, 2) + y(1, 1)$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} x + y \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & : & a \\ 2 & 1 & : & b \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 2R_1 \quad \begin{bmatrix} -1 & 1 & : & a \\ 0 & 3 & : & b+2a \end{bmatrix}$$

$$\frac{1}{3} R_2 \rightarrow R_2 \quad \begin{bmatrix} -1 & 1 & : & a \\ 0 & 1 & : & \frac{b+2a}{3} \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2 \begin{bmatrix} -1 & 0 & : & a - \frac{b+2a}{3} \\ 0 & 1 & : & \frac{b+2a}{3} \end{bmatrix}$$

$$R_1 \rightarrow (-1)R_1 \begin{bmatrix} 1 & 0 & : & -a + \frac{b+2a}{3} \\ 0 & 1 & : & \frac{b+2a}{3} \end{bmatrix}$$

$$x = -a + \frac{b+2a}{3} = \frac{-3a + b + 2a}{3} = \frac{b-a}{3}$$

$$y = \frac{b+2a}{3}$$

Verification:

$$\begin{bmatrix} a \\ b \end{bmatrix} = \left(\frac{b-a}{3}\right) \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \left(\frac{b+2a}{3}\right) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= -\left(\frac{b-a}{3}\right) + \frac{b+2a}{3}$$

$$2\left(\frac{b-a}{3}\right) + \frac{b+2a}{3}$$

$$= \begin{bmatrix} -\frac{b-a}{3} + \frac{b+2a}{3} \\ \frac{2b-2a}{3} + \frac{b+2a}{3} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{-b+a+b+2a}{3} \\ \frac{2b-2a+b+2a}{3} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{-b+a+b+2a}{3} \\ \frac{2b-2a+b+2a}{3} \end{bmatrix}$$

$$= \begin{bmatrix} 3a/3 \\ 3b/3 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

yes, these span $\mathbb{R}^2$.

Determine whether $v_1 = (1, 1, 2)$, $v_2 = (1, 0, 1)$ and $v_3 = (2, 1, 3)$ span the vector space $\mathbb{R}^3$.

Let b is an arbitrary vector $b = (b_1, b_2, b_3) \in \mathbb{R}^3$

$$b = v_1 x + v_2 y + v_3 z$$

$$(b_1, b_2, b_3) = (1, 1, 2)x + (1, 0, 1)y + (2, 1, 3)z$$

$$\begin{bmatrix} 1 & 1 & 2 & b_1 \\ 1 & 0 & 1 & b_2 \\ 2 & 1 & 3 & b_3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1 \begin{bmatrix} 1 & 1 & 2 & : & b_1 \\ 0 & -1 & -1 & : & b_2 - b_1 \\ 2 & 1 & 3 & : & b_3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1 \begin{bmatrix} 1 & 1 & 2 & : & b_1 \\ 0 & -1 & -1 & : & b_2 - b_1 \\ 0 & -1 & -1 & : & b_3 - b_1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \quad \begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & -1 & -1 & b_2 - b_1 \\ 0 & 0 & 0 & b_3 - b_1 \end{array}$$

The system is inconsistent
 V_1, V_2 and V_3 do not span R^3

Q# Determine a Spanning Set for P_n .

$\Rightarrow$ The polynomials $1, x, x^2, \dots, x^n$ span the vector space P_n .
 Since each polynomial P in P_n can be written as:

$$a_0 + a_1x + \dots + a_nx^n$$

which is a linear combination

of $1, x, x^2, \dots, x^n$

$$P_n = \text{Span}\{1, x, x^2, \dots, x^n\}$$

Q# Determine whether the polynomials $p_1 = 2+x$, $p_2 = 1-x+x^2$, $p_3 = 2+x$ span P_2 .

Let an arbitrary vector $b = b_1 + b_2x + b_3x^2 \in P_2$

$$b = \underline{P_1}x + \underline{P_2}y + \underline{P_3}z$$

$$(b_1 + b_2x + b_3x^2) = x(2+x) + y(1-x+x^2) + z(2+x)$$

$$(b_1 + b_2x + b_3x^2) = (2x + y + 2z) + (-y + z)x + (x + y)x^2$$

$$(b_1 + b_2x + b_3x^2) = (2x + y + 2z) + (-y + z)x + (x + y)x^2$$

$$\therefore 2x + y + 2z = b_1$$

$$\therefore -y + z = b_2$$

$$\therefore x + y = b_3$$

Now form augmented matrix

$$\left[\begin{array}{ccc|c} 2 & 1 & 2 & b_1 \\ 0 & -1 & 1 & b_2 \\ 1 & 1 & 0 & b_3 \end{array} \right]$$

$$= \begin{array}{ccc|ccc|ccc} 2 & -1 & 1 & 1 & 0 & 1 & +2 & 0 & -1 \\ & 1 & 0 & & 1 & 0 & & 1 & 1 \end{array}$$

$$= 2(-1) - 1(-1) + 2(1)$$

$$= -2 + 1 + 2$$

$$= 1 \neq 0 \quad \det \neq 0 \text{ so matrix is non}$$